

one pixel under the kernel is 1, it simply increases the image region or size of foreground object and reduces the noise in the image. Dilation adds a layer of pixels to both the inner and outer boundaries of regions.

This can also be used in joining the broken parts of an object. The code is given below.

```
kernel= np.ones((5,5), "uint8") #kernel
is created of 5x5
red=cv2.dilate(red, kernel)
res=cv2.bitwise_and(img, img, mask =
red)
blue=cv2.dilate(blue, kernel)
res1=cv2.bitwise_and(img, img, mask =
blue)
yellow=cv2.dilate(yellow, kernel)
res2=cv2.bitwise_and(img, img, mask =
yellow)
```

Now, contour the colours previously described. That is, differentiate each colour with a rectangular bounded line called contour. Contour can be described as a curve joining all continuous points (along the boundary) having the same colour or intensity. To implement this in the code, use cv2.findContours function as given below.

```
(_, contours, hierarchy)=cv2.
findContours(red, cv2.RETR_TREE, cv2.
CHAIN_APPROX_SIMPLE) for pic, contour in
enumerate(contours):
    area = cv2.contourArea(contour)
    if(area>300): #for removing small
    noises
    x,y,w,h = cv2.boundingRect(contour)
    img = cv2.rectangle(img, (x,y),
    (x+w,y+h), (0,0,255), 2)
    cv2.putText(img, "RED color", (x,y), cv2.
    FONT_HERSHEY_SIMPLEX, 0.7, (0,0,255))
```

There are three arguments in cv2.findContours function. First is source image, second is contour-

retrieval mode and third is contour-approximation method. Combine all the codes and run the program.

## Testing on Windows

Install OpenCV on Windows. To do so, download OpenCV\_3.0 from <http://opencv.org/downloads.html> and Python from <https://www.python.org/download/releases/2.7/>

Then, download Numpy from <https://sourceforge.net/projects/numpy/files/NumPy/1.9.1/numpy-1.9.1-win32-superpack-python2.7.exe/download>

After installing the required software, follow the steps given below:

1. Open OpenCV folder and browse to build/python/2.7/x86. You will find a cv2.pyd file.
2. Copy cv2.pyd file.
3. Open Python folder, browse to Python2.7/Lib/site-packages and paste cv2.pyd file here.
4. Open Python (IDLE).
5. Check whether OpenCV is installed, by simply typing a command against the prompt as given below.

```
>>>import cv2
>>>
```

If you get '> >' output without any error message, that means you have successfully installed OpenCV.

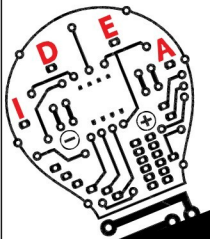
6. Double-click on multiple\_color\_tracking.py file and you will get a new window. This is the live window of the webcam. Place the object in front of the webcam or laptop camera. The program will automatically detect the colour of the object. Respective colours of the object being detected will be indicated by rectangular blocks as shown in Fig. 2.



Aquib Javed Khan (left) is a freelance technical writer. His interests include computer vision and mechatronics systems. Manjeet Kumar (right) is co-founder of Fabonix Technologies Pvt Ltd. He enjoys computer programming and debugging

# Original, innovative circuit ideas are invited

from designers and  
experimenters  
for publication in EFY magazine



Honorarium (minimum ₹1,000)  
will be paid to the contributors soon  
after publication of their articles.

Complete details may be sent to the  
'Editor, Electronics For You,  
D-87/1, Okhla Industrial Area Phase 1,  
New Delhi 110020'  
or emailed to [editsec@efy.in](mailto:editsec@efy.in)

For editorial guidelines, visit  
[http://efymag.com/Guidelines\\_for\\_EFY\\_authors.doc](http://efymag.com/Guidelines_for_EFY_authors.doc)

## EFY Note

The source code of  
this project is  
included in this  
month's EFY DVD and  
is also available for free  
download at [source.  
efymag.com](http://source.efymag.com)